

$$\begin{aligned} & -\frac{1}{2} \rightarrow \frac{1}{16\pi^2} \left(\frac{1}{72} \frac{1}{m_b^2} s_\gamma^2 y_e^{\text{pi}2} (5 g_1^2 \overline{y_e^{\text{pi}1}} + 6 c_\gamma^2 \overline{y_e^{\text{ps}}} \overline{y_e^{\text{ri}1}} y_e^{\text{rs}}) - \right. \\ & \frac{2}{27} \sum_p g_1^4 \text{LF}_{3,0} [m_d^{\text{p}}] \delta_{i1i2} + \frac{5}{36} \sum_p g_1^4 \text{LF}_{4,-1} [m_d^{\text{p}}] \delta_{i1i2} - \frac{8}{135} \sum_p g_1^4 \text{LF}_{5,-2} [m_d^{\text{p}}] \delta_{i1i2} - \\ & \frac{2}{9} \sum_p g_1^4 \text{LF}_{3,0} [m_e^{\text{p}}] \delta_{i1i2} + \frac{5}{12} \sum_p g_1^4 \text{LF}_{4,-1} [m_e^{\text{p}}] \delta_{i1i2} - \frac{8}{45} \sum_p g_1^4 \text{LF}_{5,-2} [m_e^{\text{p}}] \delta_{i1i2} - \\ & \frac{1}{9} \sum_p g_1^4 \text{LF}_{3,0} [m_l^{\text{p}}] \delta_{i1i2} + \frac{5}{24} \sum_p g_1^4 \text{LF}_{4,-1} [m_l^{\text{p}}] \delta_{i1i2} - \frac{8}{45} \sum_p g_1^4 \text{LF}_{5,-2} [m_l^{\text{p}}] \delta_{i1i2} - \\ & \frac{1}{27} \sum_p g_1^4 \text{LF}_{3,0} [m_q^{\text{p}}] \delta_{i1i2} + \frac{5}{72} \sum_p g_1^4 \text{LF}_{4,-1} [m_q^{\text{p}}] \delta_{i1i2} - \frac{4}{135} \sum_p g_1^4 \text{LF}_{5,-2} [m_q^{\text{p}}] \delta_{i1i2} - \\ & \frac{8}{27} \sum_p g_1^4 \text{LF}_{3,0} [m_u^{\text{p}}] \delta_{i1i2} + \frac{5}{9} \sum_p g_1^4 \text{LF}_{4,-1} [m_u^{\text{p}}] \delta_{i1i2} - \frac{32}{135} \sum_p g_1^4 \text{LF}_{5,-2} [m_u^{\text{p}}] \delta_{i1i2} + \\ & \frac{1}{6} s_\gamma^2 y_e^{\text{pi}2} (g_1^2 \overline{y_e^{\text{pi}1}} + 3 c_\gamma^2 \overline{y_e^{\text{ps}}} \overline{y_e^{\text{ri}1}} y_e^{\text{rs}}) \text{LF}_{1,2} [m_\Phi] + \frac{1}{12} g_1^2 s_\gamma^2 \overline{y_e^{\text{pi}1}} y_e^{\text{pi}2} \text{LF}_{2,1} [m_\Phi] - \\ & \frac{1}{36} g_1^2 (3 s_\gamma^2 \overline{y_e^{\text{pi}1}} y_e^{\text{pi}2} + 4 g_1^2 \delta_{i1i2}) \text{LF}_{3,0} [m_\Phi] + \frac{5}{24} g_1^4 \text{LF}_{4,-1} [m_\Phi] \delta_{i1i2} - \\ & \frac{4}{45} g_1^4 \text{LF}_{5,-2} [m_\Phi] \delta_{i1i2} - \frac{1}{9} g_1^4 \text{LF}_{3,0} [\tilde{\mu}] \delta_{i1i2} - \frac{1}{6} g_1^4 \text{LF}_{4,-1} [\tilde{\mu}] \delta_{i1i2} + \\ & \frac{8}{45} g_1^4 \text{LF}_{5,-2} [\tilde{\mu}] \delta_{i1i2} - \frac{1}{12} g_1^4 \text{LF}_{2,1,0} [m_1, m_e^{\text{i}1}] \delta_{i1i2} - \frac{1}{12} g_1^4 \text{LF}_{2,2,-1} [m_1, m_e^{\text{i}1}] \delta_{i1i2} + \\ & \frac{1}{6} g_1^4 \text{LF}_{3,1,-1} [m_1, m_e^{\text{i}1}] \delta_{i1i2} - \frac{1}{12} g_1^4 \text{LF}_{4,1,-2} [m_1, m_e^{\text{i}1}] \delta_{i1i2} - \frac{1}{12} g_1^4 \text{LF}_{2,1,0} [m_1, m_e^{\text{i}2}] \delta_{i1i2} - \\ & \frac{1}{12} g_1^4 \text{LF}_{2,2,-1} [m_1, m_e^{\text{i}2}] \delta_{i1i2} + \frac{1}{6} g_1^4 \text{LF}_{3,1,-1} [m_1, m_e^{\text{i}2}] \delta_{i1i2} - \frac{1}{12} g_1^4 \text{LF}_{4,1,-2} [m_1, m_e^{\text{i}2}] \delta_{i1i2} - \\ & \frac{1}{4} g_1^2 c_\gamma^2 \overline{y_e^{\text{pi}1}} y_e^{\text{pi}2} \text{LF}_{2,1,0} [m_1, m_l^{\text{p}}] + \frac{1}{2} g_1^2 c_\gamma^2 \overline{y_e^{\text{pi}1}} y_e^{\text{pi}2} \text{LF}_{3,1,-1} [m_1, m_l^{\text{p}}] - \\ & \frac{1}{4} g_1^2 c_\gamma^2 \overline{y_e^{\text{pi}1}} y_e^{\text{pi}2} \text{LF}_{4,1,-2} [m_1, m_l^{\text{p}}] - \frac{3}{4} g_2^2 c_\gamma^2 \overline{y_e^{\text{pi}1}} y_e^{\text{pi}2} \text{LF}_{2,1,0} [m_2, m_l^{\text{p}}] + \\ & \frac{3}{2} g_2^2 c_\gamma^2 \overline{y_e^{\text{pi}1}} y_e^{\text{pi}2} \text{LF}_{3,1,-1} [m_2, m_l^{\text{p}}] - \frac{3}{4} g_2^2 c_\gamma^2 \overline{y_e^{\text{pi}1}} y_e^{\text{pi}2} \text{LF}_{4,1,-2} [m_2, m_l^{\text{p}}] - \\ & \frac{1}{3} g_1^2 (c_\gamma \overline{a_d^{\text{pr}}} - s_\gamma \tilde{\mu} \overline{y_d^{\text{pr}}}) (c_\gamma a_d^{\text{pr}} - s_\gamma \tilde{\mu} y_d^{\text{pr}}) \text{LF}_{2,2,0} [m_d^{\text{r}}, m_q^{\text{p}}] \delta_{i1i2} + \\ & \frac{1}{6} g_1^2 (c_\gamma \overline{a_d^{\text{pr}}} - s_\gamma \tilde{\mu} \overline{y_d^{\text{pr}}}) (c_\gamma a_d^{\text{pr}} - s_\gamma \tilde{\mu} y_d^{\text{pr}}) \text{LF}_{3,2,-1} [m_d^{\text{r}}, m_q^{\text{p}}] \delta_{i1i2} - \\ & \frac{1}{3} g_1^2 (c_\gamma \overline{a_e^{\text{pr}}} - s_\gamma \tilde{\mu} \overline{y_e^{\text{pr}}}) (c_\gamma a_e^{\text{pr}} - s_\gamma \tilde{\mu} y_e^{\text{pr}}) \text{LF}_{2,2,0} [m_e^{\text{r}}, m_l^{\text{p}}] \delta_{i1i2} + \\ & \frac{1}{6} g_1^2 (c_\gamma \overline{a_e^{\text{pr}}} - s_\gamma \tilde{\mu} \overline{y_e^{\text{pr}}}) (c_\gamma a_e^{\text{pr}} - s_\gamma \tilde{\mu} y_e^{\text{pr}}) \text{LF}_{3,2,-1} [m_e^{\text{r}}, m_l^{\text{p}}] \delta_{i1i2} + \\ & \frac{1}{6} g_1^4 \text{LF}_{2,1,0} [m_e^{\text{i}1}, m_1] \delta_{i1i2} - \frac{1}{12} g_1^4 \text{LF}_{3,1,-1} [m_e^{\text{i}1}, m_1] \delta_{i1i2} + \\ & \frac{1}{6} g_1^4 \text{LF}_{2,1,0} [m_e^{\text{i}2}, m_1] \delta_{i1i2} - \frac{1}{12} g_1^4 \text{LF}_{3,1,-1} [m_e^{\text{i}2}, m_1] \delta_{i1i2} + \\ & \frac{1}{3} g_1^2 (c_\gamma \overline{a_e^{\text{pr}}} - s_\gamma \tilde{\mu} \overline{y_e^{\text{pr}}}) (c_\gamma a_e^{\text{pr}} - s_\gamma \tilde{\mu} y_e^{\text{pr}}) \text{LF}_{3,1,0} [m_l^{\text{p}}, m_e^{\text{r}}] \delta_{i1i2} + \\ & \frac{1}{3} g_1^2 (c_\gamma \overline{a_e^{\text{pr}}} - s_\gamma \tilde{\mu} \overline{y_e^{\text{pr}}}) (c_\gamma a_e^{\text{pr}} - s_\gamma \tilde{\mu} y_e^{\text{pr}}) \text{LF}_{3,2,-1} [m_l^{\text{p}}, m_e^{\text{r}}] \delta_{i1i2} - \\ & \frac{3}{4} g_1^2 (c_\gamma \overline{a_e^{\text{pr}}} - s_\gamma \tilde{\mu} \overline{y_e^{\text{pr}}}) (c_\gamma a_e^{\text{pr}} - s_\gamma \tilde{\mu} y_e^{\text{pr}}) \text{LF}_{4,1,-1} [m_l^{\text{p}}, m_e^{\text{r}}] \delta_{i1i2} + \\ & \frac{1}{3} g_1^2 (c_\gamma \overline{a_e^{\text{pr}}} - s_\gamma \tilde{\mu} \overline{y_e^{\text{pr}}}) (c_\gamma a_e^{\text{pr}} - s_\gamma \tilde{\mu} y_e^{\text{pr}}) \text{LF}_{5,1,-2} [m_l^{\text{p}}, m_e^{\text{r}}] \delta_{i1i2} + \\ & \frac{1}{6} g_1^2 \overline{y_e^{\text{pi}1}} y_e^{\text{pi}2} \text{LF}_{2,1,0} [m_l^{\text{p}}, \tilde{\mu}] - \frac{1}{12} g_1^2 \overline{y_e^{\text{pi}1}} y_e^{\text{pi}2} \text{LF}_{2,2,-1} [m_l^{\text{p}}, \tilde{\mu}] - \frac{1}{12} g_1^2 \overline{y_e^{\text{pi}1}} y_e^{\text{pi}2} \\ & \text{LF}_{3,1,-1} [m_l^{\text{p}}, \tilde{\mu}] + \frac{1}{3} g_1^2 (c_\gamma \overline{a_d^{\text{pr}}} - s_\gamma \tilde{\mu} \overline{y_d^{\text{pr}}}) (c_\gamma a_d^{\text{pr}} - s_\gamma \tilde{\mu} y_d^{\text{pr}}) \text{LF}_{3,1,0} [m_q^{\text{p}}, m_d^{\text{r}}] \delta_{i1i2} + \\ & \frac{1}{3} g_1^2 (c_\gamma \overline{a_d^{\text{pr}}} - s_\gamma \tilde{\mu} \overline{y_d^{\text{pr}}}) (c_\gamma a_d^{\text{pr}} - s_\gamma \tilde{\mu} y_d^{\text{pr}}) \text{LF}_{3,2,-1} [m_q^{\text{p}}, m_d^{\text{r}}] \delta_{i1i2} - \\ & \frac{5}{4} g_1^2 (c_\gamma \overline{a_d^{\text{pr}}} - s_\gamma \tilde{\mu} \overline{y_d^{\text{pr}}}) (c_\gamma a_d^{\text{pr}} - s_\gamma \tilde{\mu} y_d^{\text{pr}}) \text{LF}_{4,1,-1} [m_q^{\text{p}}, m_d^{\text{r}}] \delta_{i1i2} + \\ & g_1^2 (c_\gamma \overline{a_d^{\text{pr}}} - s_\gamma \tilde{\mu} \overline{y_d^{\text{pr}}}) (c_\gamma a_d^{\text{pr}} - s_\gamma \tilde{\mu} y_d^{\text{pr}}) \text{LF}_{5,1,-2} [m_q^{\text{p}}, m_d^{\text{r}}] \delta_{i1i2} + \\ & \frac{1}{6} g_1^2 (s_\gamma \overline{a_u^{\text{pr}}} - \tilde{\mu} c_\gamma \overline{y_u^{\text{pr}}}) (s_\gamma a_u^{\text{pr}} - \tilde{\mu} c_\gamma y_u^{\text{pr}}) \text{LF}_{2,2,0} [m_q^{\text{p}}, m_u^{\text{r}}] \delta_{i1i2} - \\ & \frac{1}{12} g_1^2 (s_\gamma \overline{a_u^{\text{pr}}} - \tilde{\mu} c_\gamma \overline{y_u^{\text{pr}}}) ($$